Business Process Document

Barendra International

Human Hair Wig Manufacturing - End-to-End Process Architecture

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This document presents the complete business process architecture of Barendra International, a Bangladesh-based human hair wig manufacturer. It maps procurement, processing, distribution, quality control, payroll, and export operations, and serves as the foundational reference for the company's ERP implementation programme.

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Executive Summary

Barendra International is a Dhaka-headquartered human hair processing and wig manufacturing enterprise that has emerged as one of Bangladesh's first formal exporters of finished human-hair products. Operating from a head office in Uttara, Sector 6, Dhaka, and a primary processing factory in Dinajpur (North Bengal), the company orchestrates a multi-stage production model that combines centralised industrial processing with a vast network of approximately 4,000-5,000 home-based female workers. The business imports raw human hair from India, Pakistan, Uzbekistan, Tajikistan, and Myanmar, supplements supply through local district-level vendors, and exports finished wigs and hair products to buyers in the United States, Europe, China, and Africa.

The company's operations currently run on a combination of Microsoft Excel workbooks, hard-copy vouchers, and a small HR-tracking module, with files centralised on Google Drive. As the workforce has scaled beyond 5,000 people and lot-level traceability has become essential for buyer compliance and inventory valuation, the owner has initiated a programme to implement a comprehensive Enterprise Resource Planning (ERP) system. This document is the foundational business process reference that will guide that ERP implementation across five sequenced phases - beginning with accounting and stock management and culminating in export-import and letter-of-credit (LC) automation.

The business follows a distinctive two-phase production model. Phase 1 distributes washed, lot-coded raw material through a five-tier hierarchy (Production Manager -> Head Leader -> Line Leader -> Home-based Supervisor -> Worker) where female workers perform hand-tool sorting and classification in their homes. Phase 2 receives the half-processed material back at the Dhaka factory, where 100 in-house workers perform machine-assisted final production - sizing by length, re-washing, colour treatment, assembly, and packing - against size-wise pricing that ranges from approximately Tk 500/kg for 5-inch hair to Tk 90,000/kg for 30-inch hair.

Three structural characteristics define the company's risk profile and automation priorities. First, the same lot number travels with the goods from raw material receipt to finished-goods storage, making lot traceability the backbone of both quality control and inventory valuation. Second, worker compensation is a hybrid of grade-based daily rates (A = Tk 90, B = Tk 75, C = Tk 50), attendance bonuses, and performance bonuses, all of which depend on daily quality grading by four Quality Controllers. Third, the home-based manufacturing model creates inherent tracking, theft, and skimming risks that the current manual system cannot fully mitigate - a single Line Controller may handle bulk payouts of approximately Tk 1.5 lakh for 100 workers, creating skimming exposure.

This document is organised into eighteen substantive sections covering company background and industry context, organisational structure, the nine core business processes (procurement, sorting and lot creation, washing, Phase 1 distribution, quality control, payroll, half-finish return, Phase 2 final production, and sales/export), inventory and lot traceability framework, consolidated business rules, pain points and operational risks, the ERP implementation roadmap, key performance indicators, and strategic implications. Each section provides process narratives, role definitions, data flows, and the specific rules that must be encoded in the future ERP system.

1. Company Background and Industry Context

1.1 Company Profile

Barendra International is a privately held human-hair processing and manufacturing enterprise headquartered in Uttara, Sector 6, Dhaka, with its primary industrial facility located in Dinajpur district in North Bengal. The company's founder and managing director, who hails from Naogaon district and holds a Sociology degree and an MBA from the University of Dhaka, entered the trade after four years of intensive market research across the United States, India, China, Nepal, Vietnam, and Malaysia, and received early trade mentoring from a Chinese business associate. The Dinajpur location was selected strategically, as the founder's wife originates from the region and operates a related hair-trade business there, providing established local supplier relationships and a recruitment base for the predominantly female home-based workforce.

The company has been in operation for approximately five to six years and positions itself as Bangladesh's first formal exporter of finished human-hair products. It maintains a representative office in India to manage raw-material sourcing relationships and coordinates with Chinese trade counterparts for finished-goods distribution. The total workforce is estimated at 4,500-5,500 people across all sites, comprising approximately 4,000-5,000 home-based female workers, 200 workers at the Dinajpur second-step factory, 100 in-house workers at the Dhaka final-production floor, and approximately 50-70 supervisory, administrative, and support staff.

1.2 Product and Material Characteristics

The company's sole raw material is real human hair - explicitly not synthetic fibre - which is processed into wigs, hair pieces, extensions, and gentlemen's caps for men. The distinction between real human hair and synthetic substitutes is commercially critical: real hair commands premium pricing (an identical product may sell to an African buyer for USD 5-50 but to an American buyer for USD 105, driven by quality perception and end-use), while synthetic substitutes can cause allergic skin reactions and are excluded from the company's product range. The company also imports specialised adhesive glue from the United States at USD 4,000-5,000 per 100 grams, a material that cannot be substituted with counterfeits without triggering customer allergies.

Two industry-specific Bengali terms describe the material states that flow through the production pipeline. Buti refers to a tightly bound, fully processed bundle that is ready for final assembly at Dhaka. Kanchi refers to a looser, longer, half-processed bundle that requires further reworking (called remik, or remake) at the Dinajpur facility before it can be promoted to the Buti state. The final wig-assembly stage is termed blended (Bengali: blended). These states - along with raw, washed, distributed, in-factory, half-finish, and finished - constitute the eight inventory buckets that the ERP system must track with real-time valuation.

1.3 Industry Context and Market Position

The global wigs and extensions market was valued at approximately USD 15.2 billion in 2025 and is projected to reach USD 31.13 billion by 2033, representing a compound annual growth rate of 9.6 percent (Grand View Research, retrieved 21 June 2026). North America accounts for approximately 45 percent of global demand, with the United States market alone growing from USD 2.79 billion in 2023 to a projected USD 6.34 billion by 2029 at 14.69 percent CAGR (Arizton). India dominates raw-hair supply, accounting for approximately 93 percent of global raw-hair exports in 2023, while China dominates finished-wig manufacturing, with the Xuchang region alone producing six of every ten wigs worldwide and employing over 300,000 workers.

Bangladesh occupies a strategically important position in this global value chain. The Indian government's Plexconcil (Ministry of Commerce) explicitly acknowledges that Myanmar and Bangladesh benefit from lower labour costs and government support, allowing them to process Indian raw hair and sell finished products competitively. Bangladesh's wig and human-hair exports reached USD 19.16 million in a single fiscal quarter according to Export Promotion Bureau data - approximately double the prior-year figure - and the Dinajpur district (particularly Chirirbandar upazila) has emerged as a recognised processing hub with 25 or more factories. The sector has grown approximately 25-fold over the past decade from a baseline of only USD 0.81 million in annual exports.

Barendra International's competitive position rests on three foundations: access to multi-country raw-material sourcing (reducing dependence on any single supplier), Bangladesh's low labour cost advantage (the national garment-sector minimum wage of BDT 12,500 per month, or approximately USD 113, is among the lowest in Asia), and the scalability of the home-based manufacturing model. However, the company faces growing buyer expectations around lot traceability and ethical sourcing substantiation - particularly from US and European buyers - which the current manual system cannot satisfy at scale. This pressure is a primary driver of the ERP initiative.

1.4 Current Systems and Automation Gap

Barendra International currently operates on Microsoft Excel workbooks and hard-copy vouchers, with files centralised on Google Drive and accessed from five to six entry-level laptops at the factory level. Two full-time data-entry operators are dedicated to transcribing the daily output of approximately 5,000 home-based workers. A small HR tracking module is used in limited fashion to monitor field supervisors' movements, but no integrated accounting, inventory, production, or payroll system exists. Internet connectivity is available at the Dhaka head office and the Dinajpur factory, with mobile hotspots used in remote field locations.

The manual system creates four categories of operational pain. First, real-time stock valuation is impossible - the owner cannot instantaneously answer basic questions such as 'how much stock do I have right now, and where is it?' Second, a single transaction currently requires multiple manual entries across cash, party, and bank ledgers, creating data-reconciliation overhead and error exposure. Third, the Asset and Liability report - a basic management instrument - takes significant time to compile manually. Fourth, lot-level traceability across the 40-plus-day production pipeline is extremely difficult to maintain in Excel, undermining both quality investigations and buyer compliance. These gaps collectively motivate the phased ERP implementation described in Section 16.

2. Organizational Structure and Roles

Barendra International's organisational structure is designed around a two-tier operational principle: centralised control over procurement, lot creation, washing, final production, and finance at the Dhaka head office, paired with decentralised execution of Phase 1 hand-processing through an extensive home-based workforce supervised by a multi-level field hierarchy. The structure scales linearly - each tier multiplies by a factor of approximately ten - which enables rapid workforce expansion but also creates the chain-of-custody and traceability challenges that the ERP system must address. The complete hierarchy is illustrated in Figure 1.

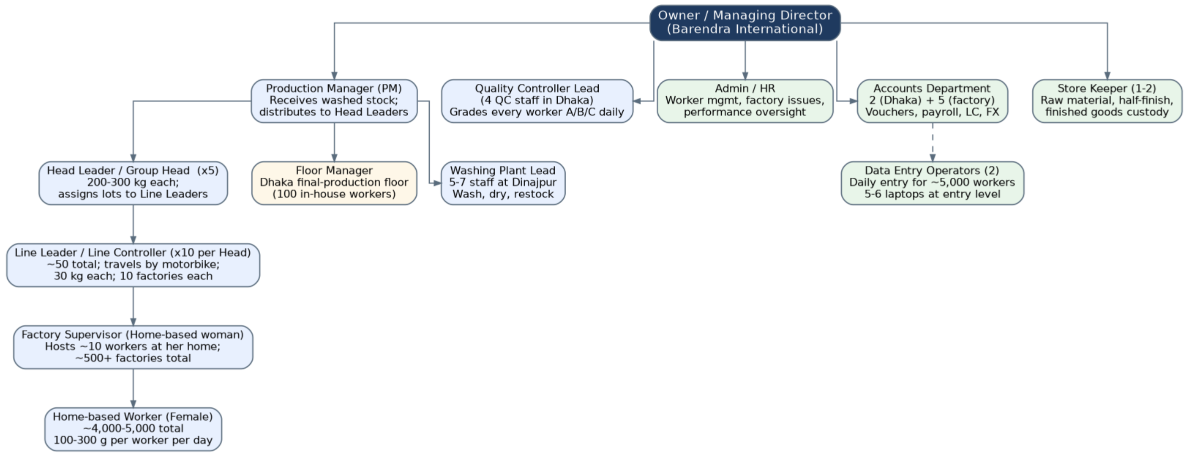

Figure 1: Barendra International organisational hierarchy and reporting lines.

2.1 Role Definitions

Role	Headcount	Reports To	Primary Responsibilities
Owner / Managing Director	1	-	Strategic decisions, final approvals, supplier relationships, key hiring
Production Manager (PM)	1	Owner	Receives washed stock, distributes to Head Leaders, oversees Phase 1 and Phase 2 production flow
Head Leader / Group Head	5	PM	Receives 200-300 kg batches; assigns lots to Line Leaders; records distribution; reviews daily reports; grades workers with QC
Line Leader / Line Controller	~50 (10 per Head)	Head Leader	Travels by motorbike; distributes 30 kg to ~10 factories; collects packets; returns reports to Head Leader; bulk salary distribution
Washing Plant Lead	1	PM	Manages 5-7 washing-plant staff at Dinajpur; oversees wash, dry, restock cycle
Floor Manager	1	PM	Supervises Dhaka final-production floor (100 in-house workers)
Factory Supervisor (Home-based)	~500+	Line Leader	Woman who hosts ~10 workers at her home; direct worker supervision; production tracking
Home-based Worker	~4,000-5,000	Supervisor	Female; performs hand-tool sorting/classification; 100-300 g per day; tagged by name+ID+lot
Quality Controller (QC)	4 (Dhaka)	Owner	Daily A/B/C grading of all workers; final QC on finished goods
Store Keeper	1-2	Owner	Custody of raw, half-finish, and finished goods; voucher management
Accountant	2 (Dhaka) + 5 (factory)	Owner	Vouchers, payments, payroll, LC, foreign-currency tracking
Admin / HR	1	Owner	Worker management, factory issues, performance oversight, recruitment
Data Entry Operator	2 + factory-level	Accountant	Daily entry of ~5,000 workers' output, weight, grade

Table 1: Complete role inventory with headcount, reporting line, and responsibilities.

2.2 Span-of-Control and Scaling Mathematics

The Phase 1 distribution hierarchy follows a deliberate 1:5:10:10:10 multiplier. One Production Manager oversees five Head Leaders, each Head Leader oversees ten Line Leaders, each Line Leader oversees approximately ten home-based factories, and each factory hosts approximately ten workers (with documented flexibility of 9, 10, or 11 workers per factory). This mathematical structure means that a single Production Manager can theoretically coordinate 5 x 10 x 10 x 10 = 5,000 workers, which closely matches the company's actual workforce. Adding capacity therefore requires adding Head Leaders (each unlocking approximately 1,000 additional workers) rather than restructuring the hierarchy.

Material allocation scales inversely with hierarchy depth. A 1,200 kg washed batch received by the Production Manager is divided into four to five Head Leader allocations of 200-300 kg each. Each Head Leader divides the allocation among ten Line Leaders at approximately 30 kg each. Each Line Leader divides the 30 kg across ten factories at approximately 3 kg per factory. Each factory supervisor divides the 3 kg among ten workers at 100-300 grams per worker per day. This granular allocation is the basis on which shortage and reject tracking is performed at every return step, and the ERP system must capture each transfer as a discrete inventory transaction with lot number, quantity, recipient, and timestamp.

2.3 Governance and Approval Authority

Approval authority is concentrated at three levels. The Production Manager approves all material distributions from the washing plant to Head Leaders and from half-finish returns into the Dhaka final production queue. Head Leaders approve daily distribution to Line Leaders and authorise worker grade entries in consultation with the Quality Controller. The Owner / Managing Director retains personal approval authority over certain stages - including supplier onboarding, large purchase orders, salary sheet final sign-off, and any out-of-tolerance shortage or reject events. This concentration of authority at the owner level is a recognised operational risk (the 'owner bottleneck') that the ERP dashboard is intended to mitigate by enabling delegated decision-making with real-time visibility.

3. End-to-End Business Process Architecture

Barendra International's operations comprise nine core business processes that together form a continuous pipeline from raw-material sourcing to finished-goods export. The pipeline runs over an extended cycle - typically 40 or more days from lot creation to finished-goods storage - because material moves through both centralised industrial stages and decentralised home-based hand-processing. Figure 2 presents the end-to-end flow, and the subsequent sections detail each process individually.

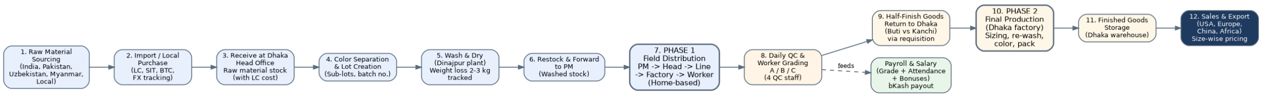

Figure 2: End-to-end business process flow from raw-material sourcing to export.

3.1 Process Inventory Summary

#	Process	Location	Primary Output	Typical Cycle
1	Raw Material Procurement	Dhaka + international	Raw hair received into Dhaka stock	Continuous
2	Initial Sorting & Lot Creation	Dhaka head office	Lot-coded, colour-separated stock	1-2 days per batch
3	Washing & Preparation	Dinajpur factory	Washed and dried restocked material	1-2 days per batch
4	Phase 1 Distribution & Field Manufacturing	Home-based factories (Dinajpur region)	Half-processed worker output	1-2 weeks per lot
5	Quality Control & Worker Grading	Dinajpur + Dhaka	A/B/C grades per worker per day	Daily
6	Salary & Payroll Management	Dhaka (payment via bKash)	Monthly salary disbursement	Monthly cycle
7	Half-Finish Return & Requisition	Dinajpur -> Dhaka	Half-finish stock at Dhaka	Per lot completion
8	Phase 2 Final Production	Dhaka factory	Finished, size-sorted goods	1-2 weeks per lot
9	Sales, Export & Order Management	Dhaka + international	Export-ready shipments	Per sales contract

Table 2: Nine core business processes with location, output, and typical cycle time.

3.2 Two-Phase Production Model

The defining architectural feature of the business is the separation of production into two distinct phases. Phase 1 encompasses all operations performed on raw or washed material through the home-based workforce - hand-tool sorting, opening, classification, and initial bundling. Phase 2 encompasses all machine-assisted and chemically intensive operations performed at the Dhaka factory - length sizing, re-washing, colour treatment, assembly, and final packing. This separation exists because the home-based environment cannot accommodate machines, chemicals, or the specialised training required for final production, while the factory environment cannot cost-effectively absorb the labour-intensive hand-sorting volume that the home-based model handles at scale.

The two-phase model creates a critical intermediate inventory state - half-finished goods - that must be tracked with the same lot identity as the original raw material. Material can spend one to two weeks in Phase 1 (distributed across dozens of home-based factories) before being returned to Dhaka for Phase 2. During this period, the ERP system must maintain visibility of every kilogram by lot, by Line Leader, by factory, and by individual worker, and must reconcile issued quantities against returned quantities with shortage and reject accounting at each level of the hierarchy.

4. Process 1 - Raw Material Procurement

4.1 Sourcing Strategy and Supplier Network

Barendra International sources raw human hair from a diversified multi-country supplier network designed to reduce dependence on any single geography and to access different hair characteristics (colour, texture, length) native to each region. International sources include India (three suppliers covering the entire country), Pakistan (one supplier), Uzbekistan, Tajikistan, and Myanmar. The company also procures locally within Bangladesh through district-level vendors - deliberately not engaging at the root collection level, which would require building an extensive grassroots collection network. A representative office in India supports direct supplier relationship management and quality inspection before shipment.

Raw-material pricing varies dramatically by quality tier. Standard import-grade hair is purchased at 1,000-2,000 taka per kilogram, while premium grades reach 12,000-14,000 taka per kilogram. Local hair purchased from Narayanganj may be approximately 2,000 taka per kilogram cheaper than equivalent Dhaka city material. The final sized-and-processed product commands radically different prices by length: 8-inch hair sells at approximately 5,000 taka per kilogram, while 30-inch hair sells at approximately 90,000 taka per kilogram - an 18-times differential that makes length retention during processing the single most important value driver in the entire pipeline.

4.2 Import Process and Documentation

International procurement is executed through Letters of Credit (LC) issued via the company's bank. The documentation workflow involves several bank-managed artefacts: SIT numbers (used on the export side and for dollar tracking), BTC numbers (bank tracking numbers), and LC documents prepared by the bank manager for the owner's signature. Multiple SIT numbers may be in flight simultaneously, each tracking pending dollar commitments, encashment status, and applicable export incentives. The bank can prepare invoices covering two to three parties at a time under a single SIT number, creating a many-to-many relationship between SIT numbers, suppliers, and LC instruments that the ERP system must model explicitly.

Foreign-currency gain and loss must be recognised on every transaction because purchases are denominated in foreign currency (primarily US dollars) while functional currency is Bangladeshi taka. The applicable bank rate on the transaction date determines the initial recorded cost, and any subsequent exchange-rate movement between commitment and encashment creates a gain or loss that must be captured automatically. The current Excel-based system handles this manually, which is time-consuming and error-prone; the ERP accounting module (Phase 1) is specifically designed to automate this recognition.

4.3 Receipt and Initial Storage

Upon arrival, raw material is received at the Dhaka head office warehouse by the Store Keeper, who records the consignment against the originating LC and SIT documentation. The material is immediately entered into the raw-material stock bucket with a valuation that includes the LC cost, landing charges, customs duty, and applicable bank charges. This initial valuation is the foundation for all subsequent cost roll-ups through the production pipeline - any error or omission at this stage propagates through to finished-goods cost and margin calculations. The ERP system must therefore enforce a complete cost-capture checklist at the goods-receipt step.

Specialised materials follow parallel procurement paths. The US-sourced adhesive glue (USD 4,000-5,000 per 100 grams) is received and stored separately with stricter custody controls because counterfeits cause customer allergies and the material cannot be substituted. Packaging materials, sizing tools, and consumables (shampoo, chemicals, dyes) are procured through standard purchase-order workflow with supplier credit terms (paona) where applicable. All procurement - whether raw material or consumable - must flow through a single unified purchasing module in the ERP to enable spend analytics and supplier performance evaluation.

5. Process 2 - Initial Sorting and Lot Creation

5.1 Colour Separation

Upon receipt and initial storage at the Dhaka warehouse, raw material enters the colour separation process. This is a manual, visually judgement-based operation in which workers physically separate hair by natural colour into distinct groupings. The industry uses a universal 1-to-10 level system for hair colour (where 1 is black and 10 is lightest blonde), but Barendra International's internal colour codes include specific identifiers such as 600 and 627, which may correspond to either colour codes or supplier lot references. Colour separation is the first quality-determining step because subsequent processing (washing, chemical treatment) is calibrated to colour characteristics, and mixing colours at this stage creates defects that cannot be economically recovered later.

5.2 Lot and Batch Number Assignment

After colour separation, each colour-separated batch is assigned a unique lot (or batch) number. Critically, lots are created based on the goods themselves - the lot is defined by the material it contains, not by an arbitrary numbering scheme. A lot and the goods it contains can be the same thing. This goods-based lot creation principle is fundamental to the entire traceability framework because it ensures that every subsequent transaction, transformation, and movement can be traced back to a specific physical batch of raw material with known colour, supplier origin, and cost characteristics.

The lot numbering system supports sub-lots: a single main lot (for example, a 120-kilogram batch) can contain multiple colour-wise sub-lots (for example, colour 600 plus colour 627) under the same main lot number. Lots can also be merged - two existing lots can be combined into a new lot - or material can be transferred from one lot to another when physical consolidation makes operational sense. These lot operations must be explicitly modelled in the ERP system as discrete transactions with audit trails, because the same physical hair may change lot identity during its lifecycle while remaining fully traceable to its original raw-material receipt.

5.3 Voucher-to-Lot Relationship

Material movement is documented through delivery vouchers, and the relationship between vouchers and lots is many-to-many. A single delivery voucher can contain multiple lots (for example, voucher 946 may contain several lots), and a single lot can span multiple vouchers as it is distributed in tranches. This many-to-many relationship is a critical data-modelling requirement for the ERP system: both the voucher entity and the lot entity must exist as first-class objects, with a junction table linking them and capturing the quantity of each lot represented in each voucher. The current Excel system cannot adequately represent this relationship, which is a primary source of traceability gaps.

5.4 Lot Lifecycle and Traceability

Once created, the lot number travels with the goods through the entire production pipeline - from Dhaka to Dinajpur, through washing, through distribution to Head Leaders and Line Leaders and factories and individual workers, back to Dhaka as half-finished goods, through Phase 2 final production, and into finished-goods storage. At every stage, the lot number is the primary key for tracking quantity, quality, cost, and timing. The ERP system must support a 'lot lifecycle view' that shows, for any lot number, the complete chain of custody with timestamps, quantities, responsible parties, and quality grades - answering questions such as 'how many days has this lot been with worker X?' or 'what is the current stock position of lot Y across all locations?' Figure 5 illustrates the complete lot lifecycle.

6. Process 3 - Washing and Preparation

After lot creation in Dhaka, material is transported to the Dinajpur factory - the company's primary industrial processing facility - for washing and preparation. The Dinajpur site houses the washing plant (staffed by 5-7 people including a floor manager and washing-plant lead), accounts staff (5 people), and 200 second-step production workers. The washing process is the first industrial transformation applied to the raw material and serves multiple purposes: it removes impurities accumulated during collection and transit, prepares the hair for colour separation consistency, and establishes the baseline weight against which all subsequent shortage and reject calculations are measured.

6.1 Wash, Dry, and Restock Cycle

The washing cycle comprises four sequential steps. First, the hair is washed using shampoo and chemical treatments calibrated to the colour and quality characteristics recorded at lot creation. Second, the washed hair is dried using controlled methods to prevent damage and to standardise moisture content. Third, the dried hair is reweighed and restocked with a new recorded quantity. Fourth, the restocked material is forwarded to the Production Manager for distribution into Phase 1 field manufacturing.

Weight loss during washing is a normal and expected outcome - typically 2-3 kilograms per batch - driven by the removal of impurities, moisture loss, and minor processing waste. This weight loss must be recorded as a formal inventory adjustment against the lot, with the new restocked quantity becoming the baseline for all subsequent distribution and reconciliation. Failure to capture this adjustment accurately creates phantom shortages downstream that cannot be explained or resolved. The ERP system must therefore enforce a weight-capture step at the end of washing, with the difference between pre-wash and post-wash quantities automatically calculated and posted to a washing-loss account.

6.2 Handoff to Production Manager

Once restocked, washed material is handed off to the Production Manager, who becomes the single point of accountability for Phase 1 distribution. A typical handoff batch is approximately 1,200 kilograms, which the Production Manager then divides among the five Head Leaders at 200-300 kilograms each. This handoff is a formal inventory transfer transaction in the ERP system - the material moves from the 'washed stock' bucket (held by the washing plant) to the 'distributed' bucket (held by the Production Manager), with the lot number preserved and the quantity, timestamp, and responsible party recorded. The Production Manager's stock position is then the basis for the next-level distribution to Head Leaders.

7. Process 4 - Phase 1 Distribution and Field Manufacturing

Phase 1 distribution is the operational heart of Barendra International's manufacturing model and the process that most differentiates it from conventional factory-based production. In this phase, washed and lot-coded material is distributed through a five-tier hierarchy to approximately 4,000-5,000 home-based female workers who perform hand-tool sorting, opening, and classification in their own homes. The workers gather at a supervisor's home (typically 5-6 adjacent homes consolidate into one work location) during a 6:30 AM to 1:30 PM work window, and each worker processes 100-300 grams of hair per day. The complete distribution and return flow is illustrated in Figure 3.

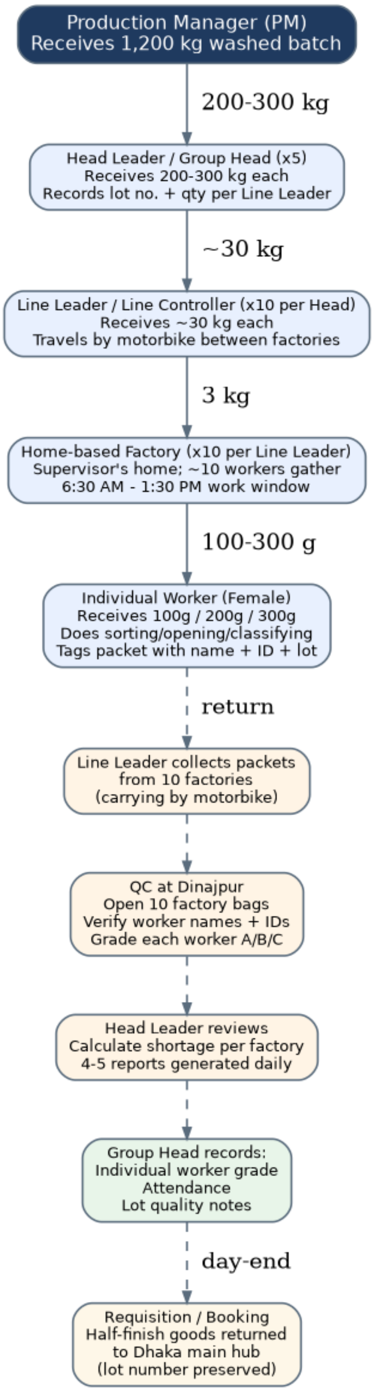

Figure 3: Phase 1 distribution and field manufacturing flow with daily return cycle.

7.1 Tier-by-Tier Distribution Mechanics

Distribution begins when the Production Manager allocates a 1,200 kg washed batch across five Head Leaders at 200-300 kg each. Each Head Leader then allocates their share across approximately ten Line Leaders at roughly 30 kg each, recording the lot number and quantity assigned to each Line Leader. Line Leaders travel by motorbike between the home-based factories in their territory - a deliberate design choice because the young men who serve as Line Leaders cannot realistically walk the distances involved, and motorbikes enable daily coverage of 10-15 factories per Line Leader. Each Line Leader distributes approximately 3 kg to each of ten factories, and each factory supervisor allocates 100-300 grams to each of approximately ten workers.

At every transfer point, two copies of a slip are created - one retained by the issuing party and one accompanying the material. The slip records the lot number, quantity, recipient name and ID, and date. This dual-copy slip system is the current chain-of-custody mechanism and will be replaced by ERP-issued digital vouchers with the same information but with automatic lot traceability and real-time stock updates. The slip system also enables the day-end reconciliation process: when workers return their processed packets, the supervisor records each packet against the issuing slip, creating a closed-loop audit trail from issue to return.

7.2 Home-Based Factory Operations

Each home-based factory is hosted by a female supervisor at her residence, where approximately ten female workers gather daily to process hair. The work uses hand tools only - no machines are permitted in the home environment - and consists of opening raw bundles, sorting by quality and length characteristics, classifying into categories, and re-bundling the processed output. Each worker's daily output is packaged individually with a slip bearing her name, worker ID, factory ID, and lot number, ensuring that quality performance can be attributed to the individual worker rather than aggregated at factory level.

The hosting supervisor receives two forms of compensation for the use of her home and her supervisory labour: a fixed hosting allowance of 100-200 taka per month (sometimes up to 400 taka) and a performance extra of 150 taka if she effectively controls worker attendance and quality output. This structure incentivises the supervisor to maintain discipline and quality at the factory level, which is critical because the home environment is inherently difficult for the company to monitor directly. Each factory is also subject to a quality threshold - a minimum of 60 percent A-grade workers - below which the supervisor's performance bonus is forfeit.

7.3 Day-End Reporting and Return Cycle

At the end of each work day, the Line Leader collects processed packets from the ten factories under his jurisdiction, carrying them by motorbike to the Quality Controller's location at Dinajpur. The ten factory bags are opened in the presence of the QC, and each packet is identified by the worker's name and ID (for example, 'this one is Fatema, this one is Jhorna, this one is BrishTi'). Each worker's output is weighed and graded A, B, or C based on length retention, breakage, colour correctness, and texture. The Line Leader then returns the collected reports and any rejected material to the Head Leader, who reviews factory-level performance, calculates shortages, and consolidates the daily reports that flow to the Production Manager and ultimately to the owner's dashboard.

Reporting volume is substantial. The company generates four to five reports per day per factory across the approximately 500 factories in the network, producing approximately 2,000-2,500 daily report entries system-wide. These reports cover factory-wise production summary, individual worker performance, salary calculation inputs, costing data, and reject/shortage tracking. The current Excel-based system requires two full-time data-entry operators to transcribe this volume, and the manual process is a primary source of delay and error in management reporting. Automating this capture at the point of QC grading - ideally with integrated weighing scales - is a Phase 3 ERP priority.

7.4 Shortage, Reject, and Wastage Accounting

At every return step, issued quantity is reconciled against returned quantity, and any variance is classified as either reject (quality failure - hair that cannot be used) or shortage (unaccounted loss). Industry-standard wastage is acknowledged as unavoidable - the owner notes that '100 percent of anything cannot be achieved' - but major losses are flagged for investigation. Typical Dinajpur second-step wastage runs 100-150 grams per kilogram, and final Dhaka production wastage runs 50-100 grams per kilogram. Reject material must still be returned with weight recorded, and a recovery rate (for example, 900 grams 'tikal' or recovered per 1,000 grams input) is calculated for each lot and each factory.

8. Process 5 - Quality Control and Worker Grading

Quality control at Barendra International operates as a continuous, daily activity rather than a discrete end-of-line inspection. Four Quality Controllers based in Dhaka are responsible for grading the output of all approximately 5,000 workers every working day, producing the A/B/C grades that directly determine each worker's daily wage. This daily grading cadence is essential because worker compensation is grade-driven - an A-grade worker earns 90 taka per day while a C-grade worker earns only 50 taka - and because the home-based manufacturing model means quality cannot be assured through process controls alone. The QC function therefore serves as both a quality gate and a payroll input.

8.1 Quality Parameters and Grading Criteria

Worker output is evaluated against four primary quality parameters. Length retention is the most important: after a standardised pulling test, hair that retains greater length (indicating less breakage) receives a higher grade. Breakage is directly penalised - workers whose output shows more broken hair receive lower grades because broken hair has lower commercial value. Colour correctness is visually judged against the lot's colour specification. Texture and feel are judged by hand, an experiential assessment that the QC cannot reduce to a fully objective standard. The owner acknowledges that QC judgement is approximately - but not exactly - 70 percent accurate, which is an inherent limitation of the manual grading model.

A critical structural feature of the grading system is lot-quality inheritance: if the raw-material lot itself is of poor quality, all workers processing that lot may receive C grades regardless of their individual effort or skill. This creates a fairness tension - workers may be penalised for lot-quality issues outside their control - and the ERP system should flag lots where the grade distribution is unusually skewed toward C so that management can distinguish lot-quality effects from worker-performance effects. Without this analytical capability, worker dissatisfaction and turnover risk increase.

8.2 The A/B/C Grade System

Grade	Daily Rate (Tk)	Typical Output Benchmark	Implication
A	90	>= 100 g, high length retention, low breakage	Top performer; eligible for performance bonus
B	75	50-100 g, moderate quality	Acceptable; counted toward A+B bonus threshold
C	50	< 50 g output OR low quality	Below standard; below 50g = automatic C

Table 3: A/B/C worker grade system with daily rates and benchmarks.

8.3 Factory-Level Quality Threshold

Beyond individual worker grading, each factory (the unit of approximately ten workers under one supervisor) is evaluated against a quality threshold: a minimum of 60 percent of the factory's workers must achieve A grade in a given period. Below this threshold, the supervisor's performance bonus is forfeit. This factory-level threshold creates a supervisory accountability mechanism - the supervisor is incentivised to train, coach, and where necessary replace underperforming workers to maintain the factory's overall grade distribution. The ERP system must calculate and display the factory A-grade percentage on a rolling basis, with automatic alerts when a factory approaches or breaches the 60 percent threshold.

8.4 Reject and Shortage Handling

Reject items (hair that fails quality standards) are weighed separately from good output and from shortage (unaccounted loss). The combined reject-plus-shortage percentage is calculated for each worker, each factory, and each lot. For example, if 5 kilograms are issued to a factory and 4.9 kilograms are returned, the 100-gram variance is decomposed into reject (e.g., 10 grams at 0.2 percent) and shortage (e.g., 20 grams), with the combined loss percentage calculated for management review. Reject material is still physically returned with weight recorded - it is not discarded at the factory level - because the company may be able to recover value from lower-grade applications. The ERP system must track reject and shortage as separate inventory transaction types with distinct cost-account postings.

9. Process 6 - Salary and Payroll Management

Barendra International's payroll system is a hybrid model that combines grade-based daily rates, attendance bonuses, and performance bonuses to align worker compensation with both output quality and reliability. The system applies to all home-based workers on a daily-labour (not monthly-salary) basis, with monthly disbursement via bKash mobile payment. The complete salary calculation flow is illustrated in Figure 6, and the detailed components are described below. The payroll process is currently manual and consumes significant management time, making it a Phase 3 priority for ERP automation.

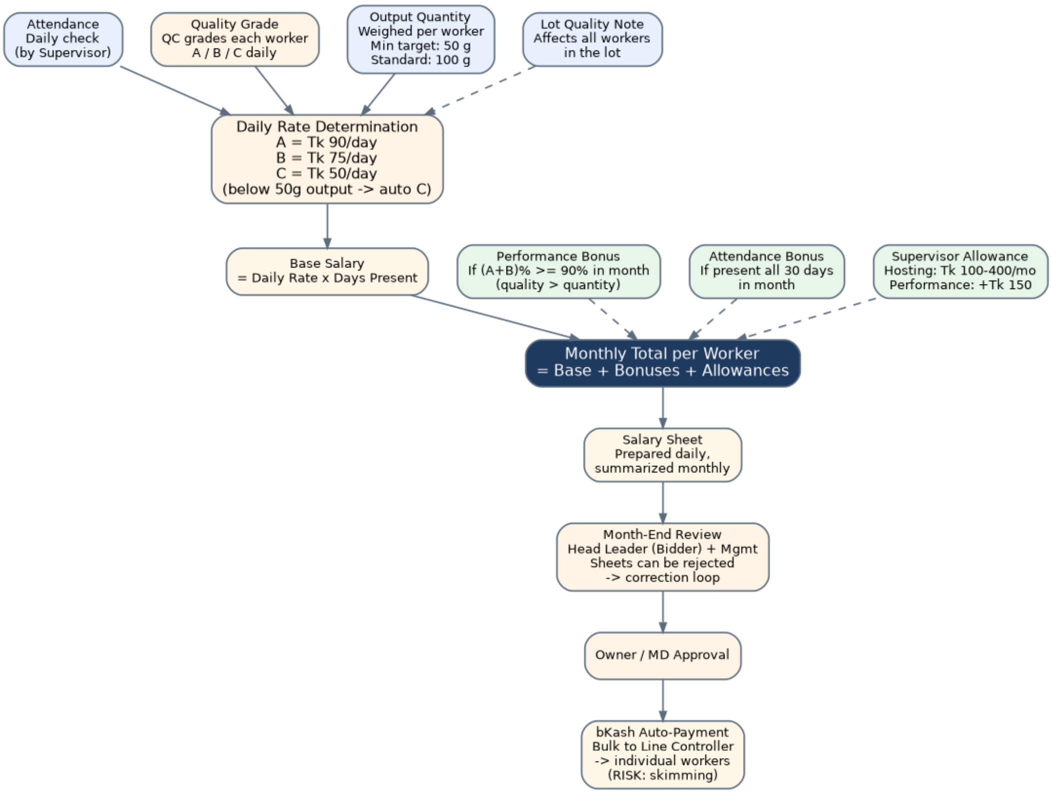

Figure 4: Salary calculation flow from daily inputs through bKash disbursement.

9.1 Compensation Components

Component	Basis	Amount / Trigger	Recipients
Daily Base Rate - Grade A	QC grade	Tk 90 per day	A-grade workers
Daily Base Rate - Grade B	QC grade	Tk 75 per day	B-grade workers
Daily Base Rate - Grade C	QC grade / below 50g	Tk 50 per day	C-grade workers
Performance Bonus	Monthly quality	Triggered when (A+B)% >= 90%	Eligible workers
Attendance Bonus	Monthly attendance	Triggered when present all 30 days	Eligible workers
Supervisor Hosting Allowance	Monthly fixed	Tk 100-400 per month	Home-based supervisors
Supervisor Performance Extra	Monthly quality	+Tk 150 if controls workers well	Eligible supervisors
Line Leader Salary	Fixed monthly	Fixed amount (not disclosed)	Line Leaders

Table 4: Complete compensation structure with basis, amount, and recipients.

9.2 Quality Over Quantity Principle

A defining design principle of the compensation system is that quality is rewarded over quantity. The owner explicitly states that producing 5 kilograms of mediocre output is less valuable than producing 100 grams of best-quality output, because the profit margin on high-quality hair is dramatically higher. This is why the monthly bonus is termed a 'performance bonus' rather than a 'production bonus' - the trigger is quality-grade achievement (A+B percentage exceeding 90 percent) rather than volume produced. The minimum daily output target of 50 grams exists primarily to identify and weed out workers who are not meeting basic productivity, with sub-50-gram output triggering an automatic C grade regardless of quality.

9.3 Salary Sheet Approval Workflow

The salary preparation workflow begins with daily summary sheets compiled by the data-entry operators from QC grading inputs. These daily sheets are aggregated into a monthly salary sheet for each Line Leader's territory. At month-end, the Head Leader (referred to as the 'bidder' in this context) sits with management to review the sheets before billing. The review is a quality gate - sheets can be rejected and returned for correction if errors are found, triggering a rework cycle. Only after the bidder and management concur does the sheet proceed to the owner for final approval, after which the bKash disbursement is initiated.

9.4 bKash Disbursement and Skimming Risk

Salary disbursement currently flows through bKash mobile payment, but the disbursement model introduces a structural skimming risk. The approved monthly payout for a single Line Leader's territory (approximately 100 workers) can total approximately 1.5 lakh taka (150,000 taka). This bulk amount is transferred to the Line Leader's bKash account, who is then responsible for distributing the correct individual amounts to each worker. The risk - explicitly acknowledged by the owner - is that the Line Leader may take more for himself and less for poorer workers, or may delay disbursement. The ERP system must address this by enabling direct bKash disbursement to individual worker accounts from the approved payroll, eliminating the Line Leader as an intermediary in the payment flow while preserving his role in distribution and supervision.

9.5 Costing Per Worker, Per Factory, Per Kilogram

Beyond individual payroll, the company tracks costing at three levels of granularity. Per-worker costing combines the worker's wages, allocated overhead, and input material cost to produce a unit cost per gram of output. Per-factory costing aggregates all workers in a factory plus the supervisor allowance, fuel costs for the Line Leader's motorbike visits, and transport costs - typical daily factory cost ranges from 1,500 taka (low day) to 2,478 taka (high day), with a 30-day average of approximately 1,632 taka per factory per day. Per-kilogram costing is the most strategic metric: for example, 60 workers producing 162 kilograms in a day yields a cost of approximately 310 taka per kilogram, which is then compared against the size-wise sale price to determine margin.

10. Process 7 - Half-Finish Return and Requisition

When Phase 1 home-based processing is complete for a given lot, the half-processed material must be returned from the Dinajpur field operations to the Dhaka main factory for Phase 2 final production. This return is managed through a formal requisition and booking process that preserves the original lot number end-to-end and transfers inventory custody from the field hierarchy back to the Dhaka store keeper. The return flow is a critical inventory transaction because it converts material from the 'in-factory' bucket (held by home-based factories) to the 'half-finish return' bucket (held by Dhaka), and any discrepancy between issued and returned quantities must be fully reconciled and explained.

10.1 Buti versus Kanchi Material States

At the half-finish stage, material is classified into one of two states that determine its next processing destination. Buti refers to tightly bound, fully processed bundles that are ready for Phase 2 final production at Dhaka - these are shipped directly to the Dhaka factory. Kanchi refers to looser, longer, half-processed bundles that require further reworking (remik) at the Dinajpur facility before they can be promoted to the Buti state and shipped to Dhaka. This state-based routing means that not all material from a given lot necessarily travels to Dhaka simultaneously - some may cycle through the Dinajpur remik process first, and the ERP system must track Buti and Kanchi as distinct sub-states within the half-finish inventory bucket.

10.2 Requisition and Booking Workflow

The return workflow is initiated by the Group Head (Head Leader), who raises a requisition to transfer completed half-finish goods from the field to Dhaka. The requisition references the original lot number and specifies the quantity, material state (Buti or Kanchi), and destination. The material is then 'booked' - removed from the Dinajpur field stock position and assigned a booking reference that travels with the physical consignment. Transport is handled by the company's own vehicles rather than courier services, because the high value-to-weight ratio of processed hair (a single lot can be worth approximately 1 lakh taka for 4 kilograms) makes third-party courier risk unacceptable.

Upon arrival at Dhaka, the store keeper receives the consignment against the original lot number, performs a quantity verification, and posts the receipt to the 'half-finish return' inventory bucket. The lot's status is updated to reflect that it has completed Phase 1 and is now available for Phase 2 final production. The ERP system must support a 'lot status' state machine that tracks each lot through the stages of: created, washing, distributed, in-factory, half-finish-returned, in-final-production, and finished - with timestamps and responsible-party capture at each transition.

11. Process 8 - Phase 2 Final Production

Phase 2 final production is performed at the Dhaka factory by approximately 100 in-house workers under the supervision of a Floor Manager, with quality oversight from the four Dhaka-based Quality Controllers. Unlike the home-based Phase 1 work, Phase 2 uses machines and chemicals that cannot be operated safely or economically in a home environment, and the workers here are trained specialists earning higher wages than their home-based counterparts. The Phase 2 workflow transforms half-finish material into size-sorted, colour-treated, finished goods ready for export, and the entire flow is illustrated in Figure 5.

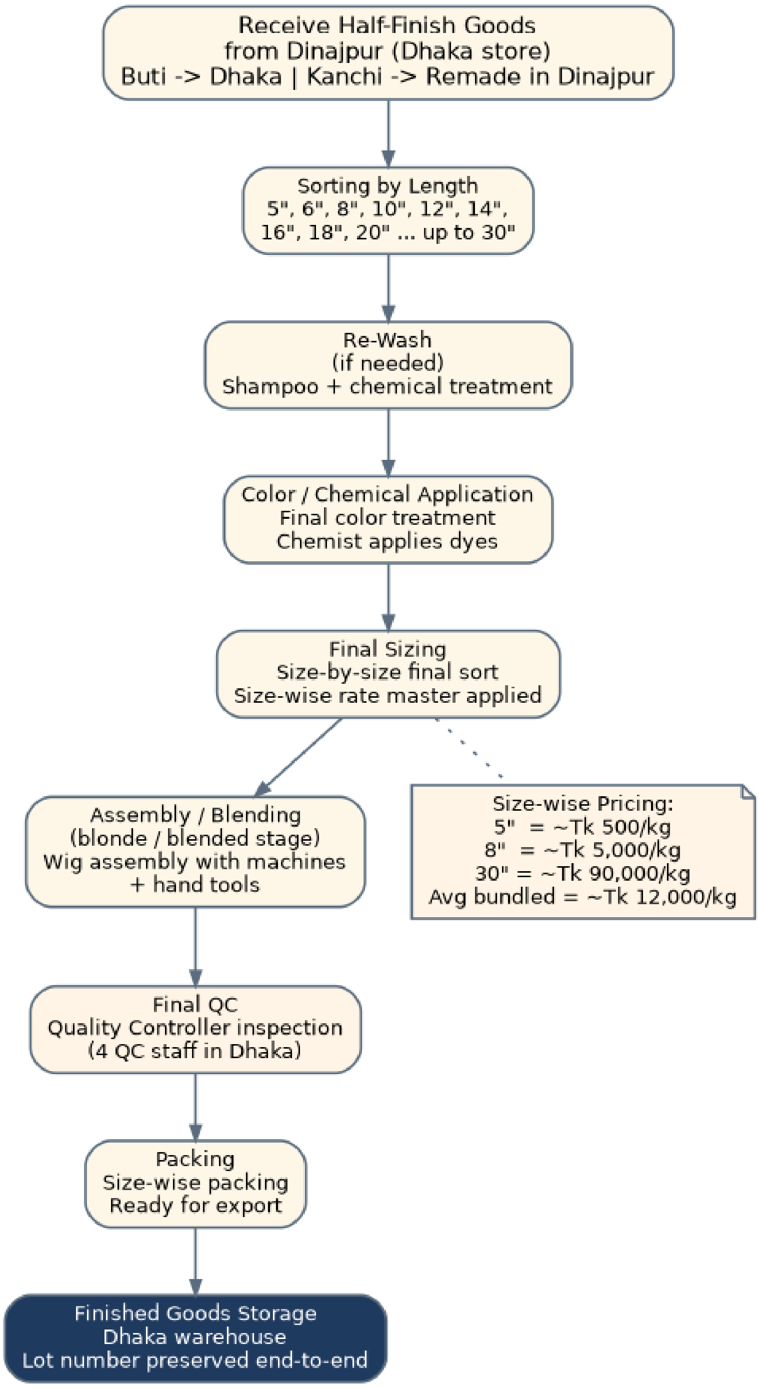

Figure 5: Phase 2 final production flow at Dhaka factory with size-wise pricing.

11.1 Sizing - The Critical Value Driver

The first and most consequential operation in Phase 2 is length sizing. Half-finish material is sorted into discrete length categories - 5-inch, 6-inch, 8-inch, 10-inch, 12-inch, 14-inch, 16-inch, 18-inch, 20-inch, and up to 30-inch - because the sale price per kilogram is exquisitely sensitive to length. The price differential is dramatic: 5-inch hair sells at approximately 500 taka per kilogram, 8-inch hair at approximately 5,000 taka per kilogram, and 30-inch hair at approximately 90,000 taka per kilogram. This pricing structure means that length retention throughout Phase 1 (where breakage reduces average length) is directly correlated with revenue, and that accurate sizing in Phase 2 is essential for correct invoicing and margin calculation.

The sizing operation is performed in full - the company does not selectively process only small or only large sizes from a given lot. A single lot's output is distributed across the full range of size categories, with each size receiving its own rate from a size-wise rate master. The blended average cost of a sized lot may be approximately 12,000 taka per kilogram. The ERP system must maintain this size-wise rate master as a configurable master data table, must record the sized quantity in each category per lot, and must calculate the lot's total realisable value as the sum of (quantity in size category x rate for that category) across all size categories produced.

11.2 Re-Wash, Colour Treatment, and Assembly

After sizing, material may be re-washed if quality inspection indicates residual impurities from Phase 1. The re-wash uses the same shampoo and chemical treatment calibrated to the sized material's characteristics. Following re-wash, the chemist applies final colour treatment - dyes and chemical processes that produce the finished colour specification required by the buyer. This is a precision operation because colour mismatch is a primary cause of buyer rejection. After colour treatment, material proceeds to final sizing (a second size-by-size sort to verify post-colour dimensions), then to assembly (the 'blended' stage where the wig or hair piece is actually constructed using machines and hand tools), and finally to QC inspection.

11.3 Final QC, Packing, and Storage

The Final Quality Controller inspects each finished item against the buyer's specification (or, for stock production, against the company's internal quality standard). Approved items proceed to packing, where they are packaged size-wise with appropriate labelling and prepared for either export shipment or finished-goods storage. Packed goods are stored in the Dhaka finished-goods warehouse with the original lot number preserved as the traceability key - even at this final stage, a finished item can be traced back through Phase 2, Phase 1, washing, lot creation, and raw-material receipt to identify the original supplier, country of origin, and purchase cost. This end-to-end traceability is increasingly demanded by US and European buyers for ethical-sourcing substantiation.

11.4 Specialised Material: Adhesive Glue

The US-imported adhesive glue (USD 4,000-5,000 per 100 grams) is consumed during the assembly stage and warrants special inventory control. The glue cannot be substituted with counterfeits because fake adhesive causes allergic reactions in end users, which would destroy the company's reputation and expose it to buyer claims. The ERP system should track adhesive glue as a separately controlled inventory item with batch-level traceability, consumption recording against specific finished-goods lots, and automatic reorder triggers when stock falls below a defined threshold. The current Excel system treats glue as a generic consumable, which is inadequate given its unit cost and quality criticality.

12. Process 9 - Sales, Export and Order Management

12.1 Market and Buyer Segments

Barendra International sells finished human-hair products to a diversified international buyer base spanning four major market segments. The United States is the highest-value market, with American buyers paying up to USD 105 for products that sell to African buyers at USD 5-50 - the differential reflecting quality tier, end-use application, and buyer negotiation power. Europe represents a second premium market with quality and ethical-sourcing expectations similar to the United States. China is both a competitive manufacturing centre and a market - the company coordinates with Chinese trade counterparts who distribute Bangladeshi-processed product, and the owner notes that 'almost all our work goes through them' in certain product lines. Africa represents a high-volume, lower-price segment, while local Bangladeshi sales and distribution across North Bengal districts supplement export revenue.

12.2 Sales Contract and Export Process

Export sales begin with a sales contract received from the buyer specifying product specifications, quantities, sizes, colours, prices, and delivery terms. The company then produces against the contract, allocating appropriate finished-goods lots to fulfil the order. Export execution mirrors the import process: LC management, SIT numbers on the export side, bank-prepared documentation, and dollar tracking of pending export receivables. Export incentives available from the bank (and from the Bangladesh Export Promotion Bureau) must be tracked and claimed. The ERP Phase 5 module - Export-Import and LC Management - is designed to automate this workflow, including document upload, SIT/BTC tracking, dollar encashment, and incentive calculation.

12.3 Pricing Strategy

Pricing is determined primarily by length (size-wise rate master) and quality tier, with buyer-specific negotiation applied on top. The same physical product may be priced very differently to different buyers based on their market position and willingness to pay. The owner cites the example of an African buyer paying USD 5-50 for a product that commands USD 105 from an American buyer - both transactions can be profitable because the underlying cost structure (raw material plus processing) is low enough to support the lower price point, while the premium buyer's higher payment reflects brand, quality assurance, and channel value. The ERP system must support buyer-specific price lists layered on top of the size-wise rate master, with margin analysis at the buyer-and-product level.

13. Inventory and Lot Traceability Framework

Inventory management at Barendra International is built on two foundational concepts: the eight-bucket stock model that tracks material through its physical transformations, and the lot-based traceability framework that maintains identity continuity from raw material receipt to finished-goods storage. These two concepts together form the data architecture that the ERP system must implement, and they represent the single most important capability gap in the current Excel-based system. Figure 7 illustrates the lot lifecycle, and Figure 8 illustrates the stock bucket model.

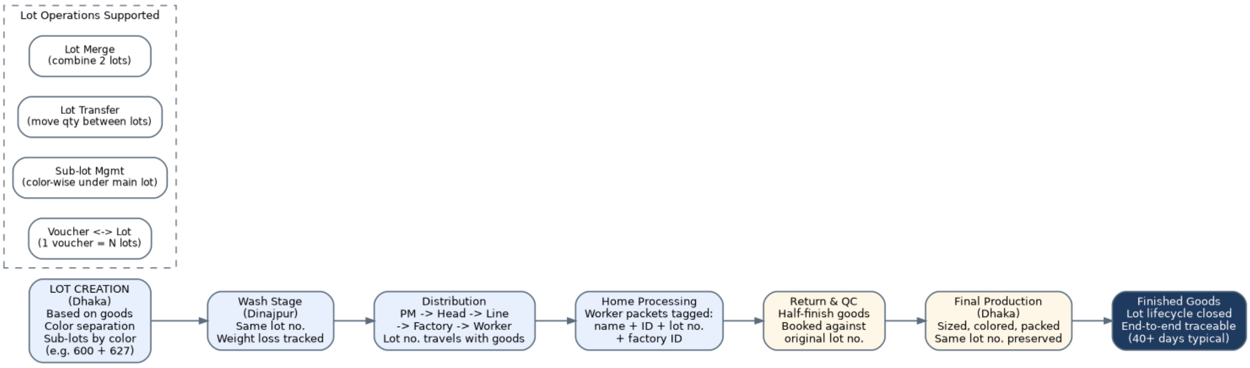

Figure 6: Lot lifecycle from creation through finished goods, with supported lot operations.

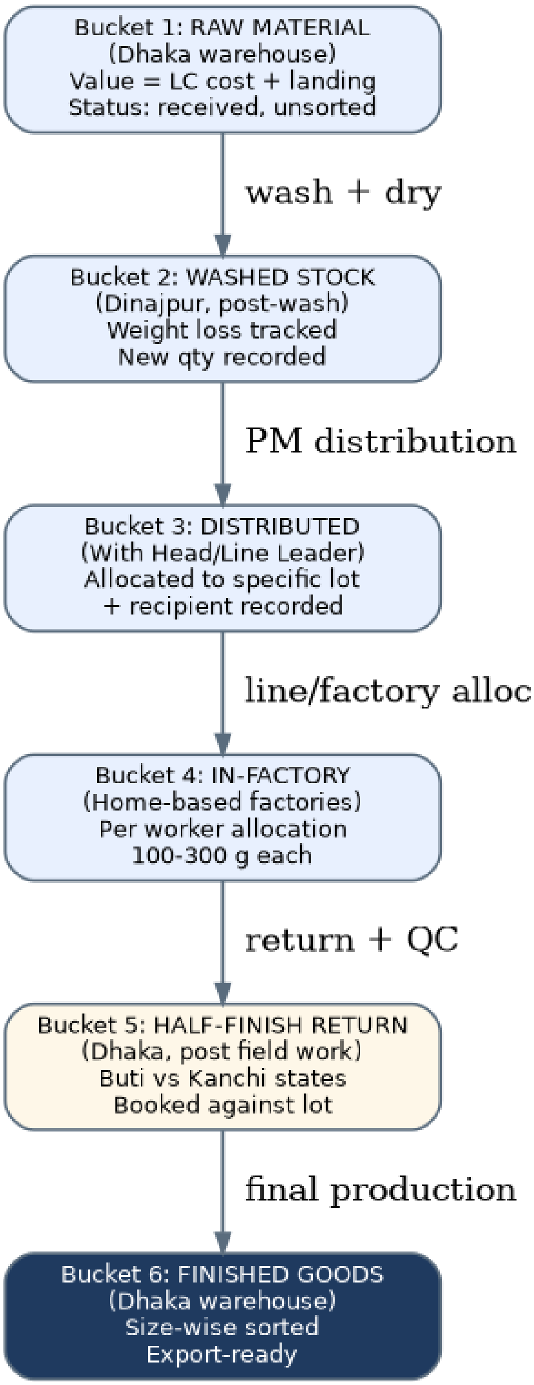

Figure 7: Eight-bucket inventory model tracking material through all transformation stages.

13.1 The Eight Inventory Buckets

#	Bucket	Custodian	Value Basis
1	Raw Material	Dhaka warehouse (Store Keeper)	LC cost + landing + duty
2	Washed Stock	Dinajpur washing plant	Raw cost + wash processing cost
3	Distributed (to Head/Line)	Production Manager -> Head Leaders -> Line Leaders	Washed cost + distribution overhead
4	In-Factory (Home-based)	Factory supervisors -> workers	Distributed cost + allocation overhead
5	Half-Finish Return	Dhaka warehouse (Store Keeper)	In-factory cost + Phase 1 processing
6	In Final Production	Dhaka factory (Floor Manager)	Half-finish cost + Phase 2 conversion
7	Finished Goods	Dhaka warehouse (Store Keeper)	Full cost roll-up + margin
8	Reject / Wastage	QC / Store Keeper	Written down or recovery value

Table 5: Eight inventory buckets with custodian and value basis.

13.2 Lot Operations and State Machine

Beyond the linear progression through the eight buckets, the lot data model must support four non-linear operations. Lot merge combines two existing lots into a new lot (for example, when small remnant quantities from two related lots are consolidated for efficient processing). Lot transfer moves quantity from one lot to another without creating a new lot (for example, when colour-wise sub-lots are reorganised under a different main lot). Sub-lot management maintains colour-wise subdivisions within a main lot. Voucher-to-lot mapping maintains the many-to-many relationship between delivery vouchers and lots. Each of these operations must be an explicit, audited transaction in the ERP system with before and after lot-quantity snapshots.

The lot itself progresses through a state machine with seven discrete states: created (at Dhaka after colour separation), washing (at Dinajpur), distributed (allocated to Head Leaders), in-factory (with home-based workers), half-finish-returned (back at Dhaka), in-final-production (Phase 2 at Dhaka factory), and finished (in finished-goods storage). Each state transition is timestamped, attributed to a responsible user, and recorded with any quantity adjustment (such as washing weight loss). The ERP system must enforce this state machine - lots cannot skip states or move backward without explicit override - and must provide a lot-lifecycle view that displays the complete state history for any lot on demand.

13.3 Real-Time Valuation Requirement

The owner's single most important reporting requirement is real-time stock valuation - the ability to see, at any moment, the total value of inventory across all locations and buckets, broken down by bucket, by lot, and by location. This requirement is currently unmet by the Excel system, where stock valuation requires manual compilation and is therefore available only periodically rather than on demand. Real-time valuation requires that every inventory transaction - receipt, distribution, transformation, return, and dispatch - immediately updates the value of the affected bucket and lot, with cost roll-ups performed automatically as material progresses through the pipeline. This is the core technical capability that justifies the ERP investment.

14. Key Business Rules and Policies

This section consolidates the twenty-five explicit business rules extracted from the meeting transcription. These rules must be encoded in the ERP system as configuration, validation logic, or workflow constraints. Each rule is stated with its enforcement point and the ERP module responsible for enforcement.

#	Business Rule	Enforcement Point	ERP Module
R1	Lots are created based on goods, not arbitrary numbering	Lot creation screen	Stock / Production
R2	One lot can have multiple colour-wise sub-lots	Sub-lot creation	Stock
R3	Lots can be merged or quantities transferred between lots	Lot operations screen	Stock
R4	One voucher can contain multiple lots (many-to-many)	Voucher entry	Stock / Accounting
R5	Same lot number travels with goods end-to-end	All transactions	All modules
R6	Distribution hierarchy: PM -> 5 Head -> 10 Line -> 10 Factory -> 10 Workers	Distribution entry	Production
R7	Worker count per factory is 9-11 (not strictly fixed at 10)	Factory master	Production
R8	Below 50g output = automatic C grade	QC grading	Worker / Payroll
R9	Each factory must have minimum 60% A-grade workers	Factory performance calc	Production / QC
R10	Supervisor hosting allowance: Tk 100-400/month	Payroll calc	Payroll
R11	Supervisor performance extra: +Tk 150 if effective	Payroll calc	Payroll
R12	Performance bonus triggered when (A+B)% >= 90%	Monthly bonus calc	Payroll
R13	Attendance bonus triggered when present all 30 days	Monthly bonus calc	Payroll
R14	Reject items must be weighed and returned with documentation	QC / return entry	Production / Stock
R15	Quality is more important than quantity (performance > production bonus)	Compensation design	Payroll
R16	Home-based work uses hand tools only (no machines)	Process design constraint	Production
R17	Machine-based final work must be done in factory, not home	Process design constraint	Production
R18	Lot quality affects all workers in that lot	Grade distribution analysis	QC / Analytics
R19	Production Manager approves all Phase 1 distributions	Approval workflow	Production
R20	Owner approves certain stages (supplier, large PO, final payroll)	Approval workflow	All
R21	Salary sheets can be rejected at month-end review	Approval workflow	Payroll

#	Business Rule	Enforcement Point	ERP Module
R22	All cash transactions must go through bank	Payment processing	Accounting
R23	Hair is high-value; lot value can be ~1 lakh Tk per ~4 kg	Security / custody	Stock / Security
R24	Foreign-currency gain/loss must be recognised on every LC transaction	FX revaluation	Accounting
R25	Real-time stock valuation is the #1 management dashboard requirement	Dashboard design	All

Table 6: Consolidated business rules with enforcement points and responsible ERP modules.

15. Pain Points and Operational Risks

The current Excel-and-hard-copy system creates operational pain across five distinct categories: manual-system overhead, tracking and theft exposure, quality and worker management challenges, reporting difficulty, and owner-centric bottleneck. Each category is analysed below with the specific pain points, their business impact, and the ERP capability that addresses them. This risk inventory directly informs the ERP implementation roadmap in Section 16.

15.1 Manual-System Overhead

The most immediate pain is the sheer volume of manual data entry required to operate at scale. Two full-time data-entry operators are dedicated to transcribing the daily output of approximately 5,000 home-based workers - a volume that consumes their full capacity and creates a backlog risk if either operator is absent. A single financial transaction currently requires multiple manual entries across cash, party, and bank ledgers to maintain consistency, multiplying the data-entry workload and creating reconciliation overhead. The Asset and Liability report - a basic management instrument - takes significant time to compile manually, which means that management decisions are made on stale data rather than current positions.

15.2 Tracking and Theft Exposure

Human hair is a 'sensitive good' - a high-value, easily stolen commodity with a value of approximately 1 lakh taka per 4 kilograms. The home-based distribution model creates inherent theft exposure: when 50 kilograms are issued to a floor controller and 49 kilograms are returned, the missing kilogram is difficult to attribute to a specific party in the chain of custody. Floor controllers may collude or 'eat the goods' (Bengali: khéya khawa), and the owner must periodically visit factories personally to deter theft - an unsustainable use of the owner's time. Competitors can also sabotage operations (Bengali: batpari) - the owner cites examples where factories have been shut down for as little as 2 lakh taka in competitor interference, through worker poaching or harassment.

15.3 Quality and Worker Management

Quality grading is judgement-based and acknowledged to be approximately 70 percent accurate, which creates fairness perceptions that can affect worker morale. Lot-quality effects may penalise workers for issues outside their control. Workers in the home environment can 'give slack' (Bengali: phanki dewa) because direct monitoring is impossible, and excessive pressure causes workers to leave - and trained workers are hard to find and expensive to replace. The hosting supervisor must be kept satisfied to maintain her cooperation in disciplining workers, creating a delicate incentive balance. Worker harassment cases (including false cases) are a legal risk that constrains management options - the owner rejected a proposed metal-detector body-scanning system precisely because of harassment-case exposure under Bangladesh law.

15.4 Owner-Centric Bottleneck

The owner is the single decision-maker for many critical operations: supplier onboarding, large purchase orders, salary sheet final approval, certain production approvals, and major exception handling. When the owner is absent - travelling, ill, or simply unavailable - the system slows or stalls. This creates a sustainability risk: the business cannot scale beyond the owner's personal bandwidth, and the owner's eventual succession or exit would leave a leadership vacuum. The ERP dashboard is explicitly intended to mitigate this bottleneck by providing delegated decision-makers with the real-time visibility they need to act within defined authority limits, and by automating approval routing for routine decisions.

15.5 Risk Register Summary

Risk	Likelihood	Impact	Mitigation (ERP Capability)
Theft in field distribution	High	High	Lot-level tracking; weight reconciliation at each transfer
Line Leader salary skimming	Medium	High	Direct bKash to worker accounts; eliminate intermediary
Worker grade disputes	Medium	Medium	Grade distribution analytics; lot-quality flagging
Inventory valuation errors	High	High	Real-time valuation; automatic cost roll-up
Lot traceability gaps	High	High	End-to-end lot tracking; state machine enforcement
Owner bottleneck	High	High	Dashboard; delegated authority; approval workflow
FX gain/loss miscalculation	Medium	Medium	Automated FX revaluation per LC transaction
Salary sheet errors	Medium	Medium	Auto-calc from QC + attendance; review workflow
Sabotage / competitor interference	Low	High	CCTV expansion; AI anomaly detection (future)
Worker shortage / turnover	Medium	Medium	Worker performance analytics; supervisor incentives
Data entry backlog	High	Medium	Auto-capture from weighing scales; mobile entry
Multi-currency reconciliation	Medium	Medium	Single-entry multi-ledger accounting automation

Table 7: Risk register with likelihood, impact, and ERP-enabled mitigations.

16. ERP Implementation Roadmap

The ERP implementation is structured as a five-phase programme designed to deliver value incrementally while managing cash-flow and change-management risk. Each phase builds on the data foundation established by the previous phase, and each phase delivers standalone business value that justifies the investment. The phasing reflects the owner's explicit priority sequence: accounting and stock first (because they underpin all financial reporting), then production (because it consumes and creates stock), then payroll (because it depends on production grading), then CRM and export-import (because they depend on finished-goods visibility). The roadmap is illustrated in Figure 8.

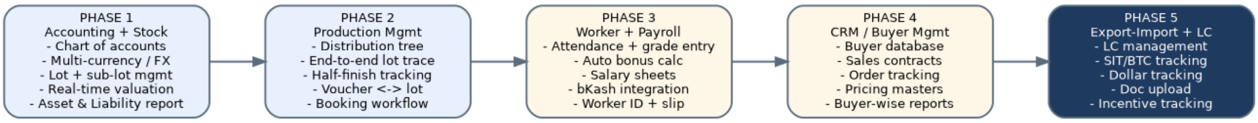

Figure 8: Five-phase ERP implementation roadmap.

16.1 Phase 1 - Accounting and Stock Management

Phase 1 establishes the financial and inventory data foundation. The accounting module implements the chart of accounts, multi-currency support with automated foreign-currency gain/loss recognition, and the single-entry-multi-ledger principle that eliminates the current duplicate data-entry overhead. The stock module implements lot and sub-lot creation, voucher-to-lot mapping, the eight-bucket inventory model, and real-time stock valuation. Phase 1 enables the Asset and Liability report to be generated on demand - the owner's most critical immediate need - and establishes the lot master data that all subsequent phases depend on. Opening balances must be loaded carefully, with particular attention to existing raw-material and half-finish stock valuation.

16.2 Phase 2 - Production Management

Phase 2 implements the full Phase 1 distribution hierarchy and the end-to-end lot traceability framework. The distribution tree (Production Manager -> Head Leader -> Line Leader -> Factory -> Worker) is modelled as a configurable hierarchy with quantity-and-lot tracking at each transfer. Half-finish goods tracking captures the return flow from field to Dhaka with Buti and Kanchi state management. The booking workflow manages requisition-based transfers between locations. Phase 2 delivers the lot-lifecycle view that enables management to answer previously unanswerable questions such as 'how many days has this lot been with worker X?' and 'what is the current stock position of lot Y across all locations?'

16.3 Phase 3 - Worker Management and Payroll

Phase 3 implements worker master data (worker ID, name, factory assignment, joining date), daily attendance capture, daily QC grade entry, and the complete salary calculation engine encompassing grade-based rates, attendance bonuses, performance bonuses, and supervisor allowances. Integration with weighing scales enables automatic weight capture at the QC grading point - the worker selects her name and factory, the scale captures the weight, the QC enters the grade, and the system posts the complete transaction in a single step. bKash integration enables direct disbursement from the approved payroll to individual worker accounts, eliminating the Line Leader intermediary and the associated skimming risk.

16.4 Phase 4 - CRM and Buyer Management

Phase 4 implements the buyer database, sales contract management, order tracking, and buyer-wise pricing masters layered on top of the size-wise rate master. Buyer-wise sales reports and buyer-specific margin analysis become available, enabling the owner to evaluate which buyers and which product mixes deliver the best margin. Phase 4 also enables forward planning - sales contracts can drive production planning, ensuring that finished goods are available when buyers require them rather than being produced speculatively and held in inventory.

16.5 Phase 5 - Export-Import and LC Management

Phase 5 completes the ERP with full export-import and LC management. LC lifecycle tracking (from issuance through encashment), SIT and BTC number management, dollar position tracking, export document upload and management, and export incentive tracking are all automated. This phase closes the loop between procurement (where LCs originate on the import side) and sales (where LCs originate on the export side), providing a complete foreign-currency position view. Phase 5 also includes asset management (tracking bikes, machines, laptops, and other fixed assets) as a parallel workstream.

16.6 Critical Cross-Cutting Features

Several features are required across all phases rather than within a single phase. The real-time owner dashboard - the #1 priority - must be available from Phase 1 onward and must progressively enrich as subsequent phases come online. Approval workflow must be configurable per phase and per transaction type. Mobile access is required for Line Leaders and field supervisors who operate in areas with only mobile-hotspot connectivity. Role-based access control must implement the distinct user roles identified in Table 1. CCTV integration with AI-based anomaly detection is a future enhancement considered for theft deterrence, contingent on legal review of privacy and harassment-case implications under Bangladesh law.

17. Key Performance Indicators and Metrics

The ERP system must surface a defined set of Key Performance Indicators (KPIs) at operational, tactical, and strategic levels. These KPIs are derived from the business rules in Section 14 and the pain points in Section 15, and they constitute the management instrument panel that will replace the current periodic manual reporting. KPIs are organised into four categories: production, quality, financial, and workforce.

Category	KPI	Frequency	Target / Threshold
Production	Daily output (kg) by factory	Daily	Per factory target
Production	Lot cycle time (creation to finished)	Per lot	< 45 days
Production	Distribution accuracy (issue vs return)	Daily	99%+
Production	Stock bucket values	Real-time	Dashboard
Quality	A-grade worker percentage (factory)	Daily	>= 60%
Quality	A+B grade percentage (worker, monthly)	Monthly	>= 90% for bonus
Quality	Reject percentage by lot	Daily	< 0.5%

Category	KPI	Frequency	Target / Threshold
Quality	Shortage percentage by factory	Daily	< 0.3%
Quality	Lot-quality skew flag (C-grade concentration)	Per lot	Investigate if > 50% C
Financial	Real-time stock valuation	Real-time	Dashboard
Financial	Asset & Liability position	Real-time	Dashboard
Financial	FX gain/loss per LC	Per transaction	Auto-tracked
Financial	Cost per kg (by factory, by lot)	Daily	< Tk 320/kg target
Financial	Margin per size category	Per shipment	Positive per size
Financial	Buyer-wise margin	Monthly	Trend analysis
Workforce	Daily attendance rate	Daily	Per factory
Workforce	Worker turnover rate	Monthly	< 5%/month
Workforce	Salary disbursement accuracy	Monthly	100% to worker accounts
Workforce	Average worker daily earning	Monthly	Tk 2,000-3,000/worker
Workforce	Supervisor performance (factory A%)	Monthly	>= 60% threshold

Table 8: KPI register across production, quality, financial, and workforce categories.

18. Conclusion and Strategic Implications

Barendra International operates a sophisticated, multi-stage human hair manufacturing business that has scaled to approximately 5,000 workers and a multi-country sourcing network on the strength of the founder's market knowledge and a well-designed two-phase production model. The business is structurally well-positioned to benefit from the global growth of the wigs and extensions market (projected at 9.6 percent CAGR through 2033) and from Bangladesh's emerging role as a recognised hair-processing and export hub. The home-based manufacturing model provides a labour cost advantage that is difficult for competitors in more developed economies to match, and the diversified multi-country sourcing network reduces supply risk.

The strategic imperative is clear and acknowledged by the owner: the business has outgrown the Excel-and-hard-copy system that supported its early growth, and continued scaling without an integrated ERP will compound the operational risks documented in Section 15. The five-phase ERP roadmap presented in Section 16 provides a pragmatic path that delivers incremental value at each phase, beginning with the accounting and stock management capabilities that will immediately enable real-time stock valuation and the on-demand Asset and Liability report that the owner most urgently needs. Each subsequent phase builds on this foundation to extend automation into production, payroll, CRM, and export management.

Three strategic implications deserve management attention beyond the ERP implementation itself. First, the owner-bottleneck risk identified in Section 15.4 is a structural sustainability concern that the ERP dashboard can mitigate but not eliminate - the company should plan for delegated authority structures and succession planning in parallel with the ERP rollout. Second, the home-based manufacturing model creates inherent theft and skimming exposure that technology can reduce but not fully close - periodic physical inventory counts, independent audit of high-value lots, and continued owner presence in the field remain necessary controls.

Third, the growing buyer demand for lot traceability and ethical-sourcing substantiation (particularly from US and European buyers) represents both a compliance risk and a competitive opportunity - the company that can demonstrate end-to-end traceability will command premium pricing and preferred-supplier status, and the ERP-enabled lot traceability framework is therefore a strategic asset, not merely an operational tool.

This document will serve as the foundational business process reference for the ERP implementation programme. It should be reviewed and updated at the completion of each ERP phase to reflect the implemented system's actual behaviour, and it should be used as the basis for user training, process documentation, and continuous improvement initiatives. The detailed business rules in Section 14, the KPI register in Section 17, and the risk register in Section 15 should be maintained as living documents that evolve with the business.

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